

MOSAIC v0.4.0 Technical Report

Model Specification and Integrated Statistical Evaluation of
Decentralized Mission Coordination among Heterogeneous Robots

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Abstract

MOSAIC is an agent-based model of decentralized mission coordination among heterogeneous robots operating under partial observability, spatially variable risk, limited energy, dynamic communication, and cooperative task requirements. The model integrates local task discovery, temporary communication links, contract-net-style auctions, bounded coalition formation, four reward regimes, reputation updating, adaptive bidding strategies, contract release, and failure traceability. This report provides a concise model specification structured according to the ODD protocol and an integrated statistical evaluation of seven hypotheses using 690 official run-level observations generated with 30 paired random seeds per condition.

Non-parametric repeated-measures methods, planned paired contrasts, Holm corrections, rank-biserial effect sizes, Kendall's W, and bootstrap intervals are used to separate robust mechanism effects from descriptive differences. The baseline achieved 92.22% mean completion (SD 6.80 percentage points), with 23 of 30 runs at or above 90%. A one-sided Wilcoxon signed-rank test supports a typical-run advantage over the 90% benchmark ($W = 231$, $p = 0.0098$, rank-biserial = 0.540), while the bootstrap 95% interval for the mean [89.67%, 94.44%] overlaps 90%; baseline viability is therefore supported with caution.

The strongest effects are structural. Increasing cooperative-task demand from 0.0 to 0.8 reduces completion by 23.22 percentage points and concentrates failures in auction/no-feasible outcomes. Moderate capability heterogeneity outperforms both homogeneous and highly heterogeneous teams. Increasing communication radius from 4 to 8 improves completion, information coverage, coalition delay, coordination overhead, and mission efficiency, whereas moderate temporary link failure has no reliable completion effect. Leader-priority rewards raise inequality and unresolved auctions without improving completion. Reputation and adaptation do not reliably improve completion, although their interaction reduces coalition delay. Mission incentives increase payouts almost perfectly with weight but do not improve completion reliably.

The integrated evidence supports a structural conclusion: MOSAIC is constrained primarily by information reach, capability compatibility, and timely coalition construction rather than by insufficient reward intensity. The highest-priority development target is decentralized partner discovery and coalition formation.

Keywords: agent-based modeling; multi-robot systems; decentralized coordination; coalition formation; task allocation; incentives; NetLogo; ODD; complex systems; simulation experiments, autonomous systems.

Executive Summary

MOSAIC asks whether incentives can convert decentralized competition among heterogeneous robots into coherent mission-level self-organization without central supervision. The evidence provides a conditional answer. Incentives influence the distribution of rewards, the congestion of auctions, and the conditional success of formed coalitions. They do not, by themselves, solve the dominant limitations in task visibility, partner reachability, capability compatibility, and coalition formation before deadlines.

The report distinguishes three sequential coordination capacities. First, information reach determines whether a task is discovered and whether its identifier reaches potentially useful robots. Second, allocation feasibility determines whether available robots can satisfy capability, energy and deadline constraints and whether a feasible coalition can be constructed. Third, execution reliability determines whether assigned robots arrive, synchronize and complete service before termination. The experimental evidence shows that allocation feasibility is the binding stage under difficult conditions.

- Cooperative-task demand is the largest tested completion driver: moving from 0.0 to 0.8 cooperative demand reduces completion by 23.22 percentage points.
- Capability heterogeneity has an inverted-U relationship with performance: the 0.5 condition exceeds the average of the 0.0 and 1.0 extremes by 5.94 percentage points.
- Communication radius is a system-level resource: radius 8 improves completion by 3.56 points, coverage by 7.36 points, coalition delay by 32.80 ticks, overhead by 22.12 units and mission efficiency by 0.332.
- Leader-priority rewards increase Reward Gini by 0.043 relative to equal-share and add 229.63 unresolved auction rounds without improving completion.
- Reputation and adaptation are active but not mission-transformative under the current baseline; their joint process benefit is a 9.65-tick reduction in coalition delay.
- Mission-incentive weights 0.25 and 0.50 are economically dominated by zero incentive on the observed cost-completion plane.

The primary engineering implication is not to increase incentive intensity. It is to improve capability-aware partner discovery, feasible coalition search, and communication-supported reach. Equal-share should remain the reference reward regime until an alternative demonstrates a measurable mission benefit or a justified fairness objective.

Table 1. Hypothesis decision register.

Hypothesis	Decision	Evidence interpretation
H1 - Baseline viability	Supported with caution	Typical-run completion exceeds 90%, but the bootstrap interval for the mean overlaps 90%.
H2 - Reward-regime trade-off	Supported	Leader-priority raises inequality and auction friction without a completion benefit.
H3 - Communication structure	Supported	Increasing radius improves mission and process outcomes; moderate temporary failure is not robust.
H4 - Heterogeneity optimum	Supported	Capability heterogeneity 0.5 outperforms 0.0 and 1.0.
H5 - Cooperative-demand threshold	Supported	High cooperative demand produces a nonlinear collapse dominated by infeasible allocation.
H6 - Reputation and adaptation	Not supported for completion; secondary evidence	No completion interaction; joint activation reduces coalition delay.
H7 - Incentive saturation	Supported	Payouts rise strongly while completion and mission efficiency do not.

1. Introduction

1.1 Research problem

Heterogeneous multi-robot missions require more than local motion control. They require distributed task discovery, communication, allocation, role compatibility, coalition construction, synchronization, resource management, incentive design and recovery after assignment failure. A central planner can optimize some of these processes when it has complete and current information. In dynamic field environments, however, complete observability, persistent connectivity and centralized availability are fragile assumptions.

MOSAIC represents the alternative coordination problem. Robots act with bounded local knowledge. They discover tasks through sensing, exchange identifiers across temporary links, submit local bids, compete for individual tasks, and form temporary coalitions for cooperative tasks. The observer schedules state transitions and collects metrics but does not construct a global mission plan, choose winners through a global optimizer or enumerate a globally optimal coalition structure.

Under what conditions can incentive mechanisms transform decentralized competition among heterogeneous robots into self-organized mission coordination without central supervision?

1.2 Analytical contribution

The report makes five contributions. First, it documents the implemented model using the Overview, Design concepts and Details structure of the ODD protocol. Second, it separates mission outcomes from information, allocation, execution, distribution and adaptation processes. Third, it reports a nine-invariant verification framework and run-level accounting residuals. Fourth, it evaluates seven mechanism-focused experiments with paired random seeds. Fifth, it integrates the findings into a design diagnosis rather than treating each experiment as an isolated comparison.

1.3 Scope

This is a technical report for a synthetic research model. It evaluates internal behaviour under specified parameter ranges. It does not estimate real-world fleet performance, validate a physical controller, certify safety, or establish economic value for a deployed system. The model has not yet been published in a public repository; references to future repository or DOI metadata are intentionally omitted from the evidential claims.

2. Model Specification Following the ODD Protocol

2.1 ODD scope and traceability

This section follows the ODD protocol structure introduced for individual- and agent-based models (Grimm et al., 2006, 2010): Overview, Design concepts and Details. It is a report-level specification that is sufficiently detailed to interpret the experiments and reproduce the implemented mechanisms. The standalone MOSAIC ODD Protocol remains the authoritative exhaustive specification of variables, procedures and formulas.

ODD compliance statement

Section 2 explicitly covers Purpose and patterns; Entities, state variables and scales; Process overview and scheduling; the eleven ODD design concepts; Initialization; Input data; and Submodels. Statistical

ODD compliance statement

methods and results are intentionally outside ODD because they concern experimental evaluation rather than model specification.

2.2 Overview

2.2.1 Purpose and intended patterns

MOSAIC is a computational laboratory for studying decentralized mission coordination among heterogeneous robots in a spatially complex and partially observable environment. Robots must discover, evaluate, bid for, travel to and execute distributed tasks. Individual tasks require one feasible robot. Cooperative tasks require a temporary coalition whose aggregate sensing, manipulation, transport and communication capabilities meet simultaneous requirements.

The intended patterns include mission completion and failure, partial task discovery, information diffusion, network isolation, contract competition, capability complementarity, coalition attempts and delay, releases, reward and contract inequality, reputation differentiation, strategy switching, coordination overhead and mission efficiency. These are model-generated analytical targets, not empirical validation claims.

2.2.2 Entities, state variables and scales

Table 2. ODD entity, state and scale summary.

Entity	Principal state variables	Role and scale
Patches	Terrain cost, hazard, communication quality, visibility, explored state, visual zone	Static 51 x 51 toroidal environment; 2,601 patches; continuous robot positions measured in patch units.
Robots	Four capabilities, speed, energy, risk tolerance, state, knowledge, contract, credits, reputation, strategy and history	15 mobile agents in the validated baseline; one active contract per robot.
Tasks	Reward, deadline, service duration, risk, capability requirements, discovery, bids, assignment, progress, release and failure	30 stationary tasks in the baseline; individual or cooperative.
Communication links	Current link quality	Undirected and temporary; destroyed and reconstructed each tick.
Global state	Information, bids, contracts, coalitions, rewards, releases, strategies and verification counters	Observer-level cumulative accounting and output reporting.
Time and units	Ticks; dimensionless energy, reward, capability, reputation and efficiency	One tick is a complete coordination-action cycle; no fixed real-time mapping.

Capabilities are bounded between 0.10 and 1.00. Reputation is bounded between 0 and 1 and begins at 0.50. Undiscovered tasks are invisible and cannot receive bids. Robot knowledge consists of persistent task identifiers; information is never forgotten. The environment is static after setup, but the communication network is dynamic.

2.2.3 Process overview and scheduling

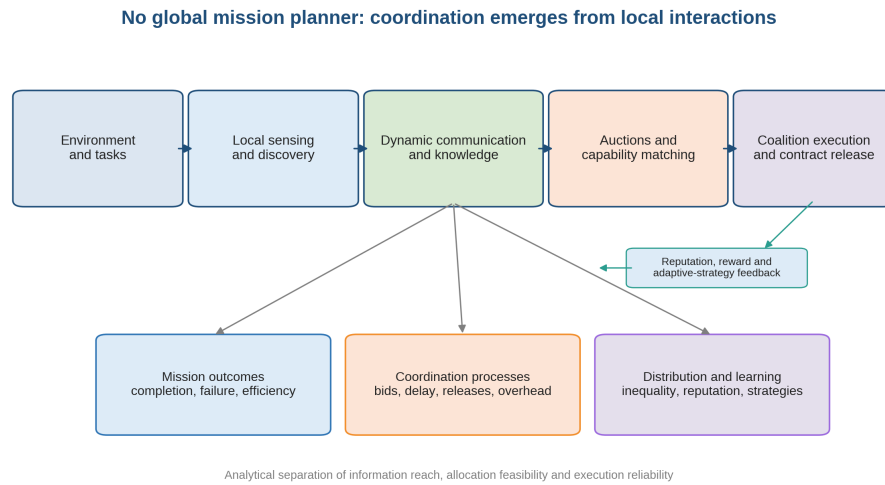


Figure 1. Conceptual architecture of the MOSAIC coordination process and analytical outcome classes.
The figure is a model interpretation, not an empirical result.

The process architecture separates information reach, allocation feasibility and execution reliability. A task must first be discovered and communicated. A robot or coalition must then satisfy capability, deadline, energy and utility constraints. Finally, assigned members must travel, synchronize and execute. Reputation, rewards and strategy values feed back into later bidding without creating a global planner.

Table 3. MOSAIC tick schedule.

Order	Procedure	Analytical consequence
1	Evaluate mission stopping conditions and task deadlines	Tasks can fail before movement or execution in the current tick.
2	Move active robots and detect nearby tasks	Discovery depends on post-movement position, sensing and visibility.
3	Destroy and rebuild communication links	Network structure reflects current positions and local communication quality.
4	Share task identifiers synchronously	Information travels at most one network edge per tick.
5	Open auctions and submit at most one bid per free robot	Bidding is bounded and knowledge-local.
6	Resolve individual contracts and cooperative coalitions	Tasks act as local auctioneers; cooperative search is greedy and bounded.
7	Validate, synchronize and execute assignments	Members must remain available, arrive and pay execution energy.
8	Release invalid contracts; update reputation and strategies	Failed coordination can reopen a task before its deadline.
9	Refresh observations and advance the tick	Metrics capture the resulting mission and process state.

2.3 Design concepts

Table 4. ODD design concepts represented in MOSAIC.

ODD concept	Implementation in MOSAIC
Basic principles	Agent-based modelling, contract-net-style allocation, distributed auctions, coalition formation, information diffusion, reputation and reinforcement-style strategy values.
Emergence	Mission completion, congestion, inequality, strategy diversity and coalition delay emerge from local sensing, communication, bidding and synchronization.
Adaptation	Risk-aware movement, reputation updates, decay toward 0.50 and epsilon-greedy strategy selection.
Objectives	A robot submits one positive-utility bid based on expected reward, urgency, reputation, strategy, energy and risk.
Learning	Only the current strategy value is updated from reward minus energy expenditure; no state model or counterfactual update.
Prediction	Approximate travel time, energy, risk, reward share, capability contribution and deadline feasibility.
Sensing	Probabilistic detection depends on sensing capability, effective radius and local visibility.
Interaction	Direct information exchange; indirect competition, coalition membership, reward allocation and reputation-sensitive bidding.
Stochasticity	Environment, agents, tasks, movement, detection, links, information transmission, ties, strategy exploration and agent order.
Collectives	Temporary capability-feasible coalitions for cooperative tasks.
Observation	Mission, coordination, information/network, distribution, adaptation and verification metrics.

2.4 Details

2.4.1 Initialization

Setup clears state, applies the random seed, initializes counters, creates a smoothed synthetic environment, samples heterogeneous robots, samples individual and cooperative tasks, constructs the first communication network, refreshes the interface and resets ticks. Patch fields are generated independently and diffused to create spatial continuity. Robots are placed away from the highest hazards and receive bounded capabilities, speed, energy and risk tolerance. Tasks occupy distinct eligible patches and receive stochastic rewards, deadlines, durations, risks and capability requirements.

Capability heterogeneity changes the dispersion of robot sensing, manipulation, transport, communication, speed and risk tolerance around central values. Cooperative tasks sample a required team size and aggregate capability requirements. All tasks begin hidden, undiscovered, unassigned and incomplete. The random seed makes the complete initialization reproducible when NetLogo version, model version and interface parameters are held constant.

2.4.2 Input data

MOSAIC uses no external input dataset, GIS layer, robot log or platform calibration. All environment, robot and task properties are generated synthetically. This design improves portability and exact seed-based replication, but it also limits empirical interpretation and prevents claims that absolute outputs represent physical units.

2.4.3 Principal submodels

Table 5. Principal model submodels.

Submodel	Operational logic	Important limitation
Exploration and movement	Correlated random walk with risk avoidance; movement consumes terrain- and hazard-dependent energy.	No path planning, collision avoidance or physical interference.
Task discovery	Unknown active tasks are detected probabilistically within a capability-scaled radius.	Task knowledge is identifier-based rather than semantic.
Dynamic communication	Pairwise links form within radius when effective environmental and capability quality exceed a threshold and survive failure.	No bandwidth, queue, routing or latency model.
Information sharing	Successful links synchronously exchange unknown task identifiers.	Knowledge never decays or becomes obsolete.
Bid utility	Expected reward, urgency, reputation and strategy adjustments minus energy and risk penalties.	Travel cost uses task-patch conditions rather than a complete future route.
Individual allocation	Highest bid wins; exact ties are random.	No global assignment optimization.
Cooperative allocation	Ranked bids are appended greedily until team size and aggregate capabilities are satisfied.	May miss feasible combinations available to combinatorial search.
Execution and synchronization	All members must arrive and retain energy; waiting or depletion can release the entire contract.	Capabilities do not change per-tick work progress once the coalition is formed.
Reward distribution	Leader-priority, equal-share, contribution-based and mission-aligned regimes.	Expected leader payment can differ from final selected leader.
Reputation and adaptation	Success/failure updates, decay, incremental strategy values and epsilon-greedy choice.	Reputation has limited signal when persistent reliability differences are weak.

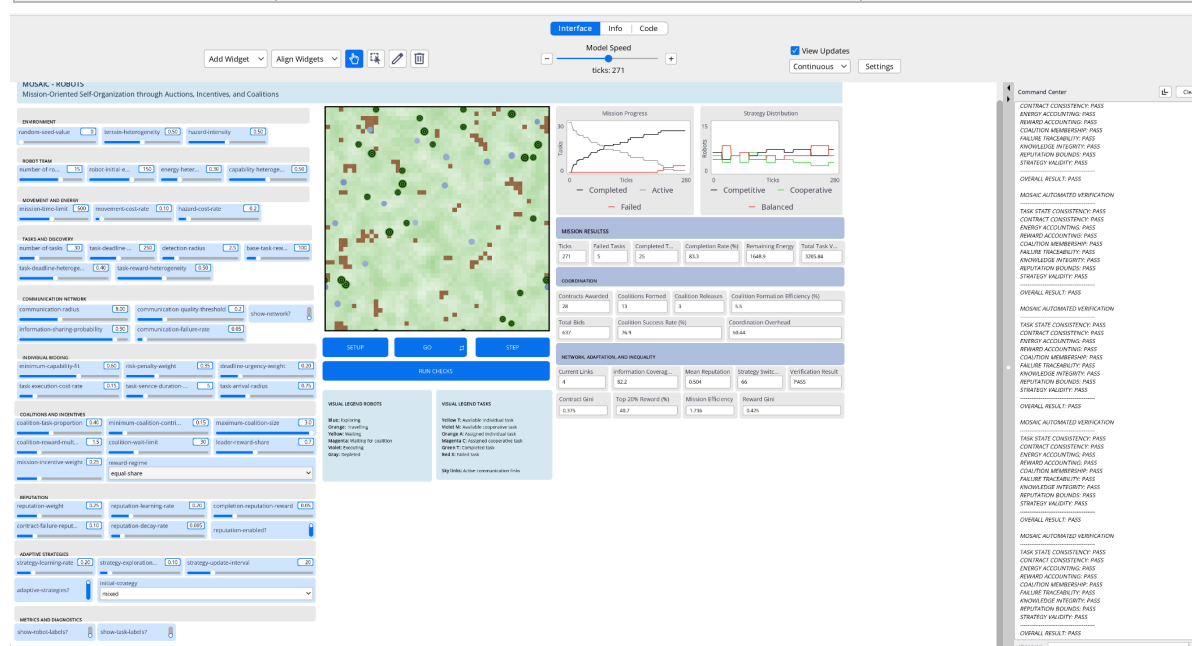


Figure 2. MOSAIC NetLogo interface during an example mission run. The interface is an operational aid and not a statistical result.

2.4.4 Observation and derived metrics

Mission outcomes include completed and failed tasks, completion rate, discovered-task completion, explored area, task value and energy use. Coordination-process outcomes include bids, contested and unresolved auction rounds, contracts, memberships, coalition attempts, formation efficiency, releases, delay and coordination overhead. Information and network outputs include degree, density, isolated robots, transmissions, unique acquisitions and coverage. Distribution and adaptation outputs include reward and contract Gini coefficients, top-reward share, reputation and strategy switching.

Coalition success and coalition formation efficiency answer different questions. Coalition success is completed cooperative tasks divided by coalitions formed; it is conditional on a coalition existing. Formation efficiency is coalitions formed divided by formation attempts; it evaluates the capacity to construct feasible teams. Coordination overhead combines bids, transmissions, coalition attempts and releases per completed task. Mission efficiency divides task value by energy and coordination activity. Both are comparative model indices rather than physical units.

3. Verification and Internal Validity

The interactive verification procedure evaluates nine invariants. It returned OVERALL RESULT: PASS for the frozen model state used for documentation. Five checks were also recorded at run level across the official experimental programme. Verification establishes consistency between state transitions and accounting identities; it does not establish empirical validity, controller safety or transferability to physical robots.

Table 6. Automated verification invariants.

Invariant	Condition checked
Task state consistency	No task is simultaneously completed and failed.
Contract consistency	Every robot contract corresponds to a task that lists that robot.
Energy accounting	Starting energy equals used energy plus remaining energy within tolerance.
Reward accounting	Robot credits equal total rewards distributed within tolerance.
Coalition membership	Every assigned task member targets that task.
Failure traceability	Every failed task has a terminal failure reason.
Knowledge integrity	Every known task identifier refers to an existing task.
Reputation bounds	Every reputation remains in $[0, 1]$.
Strategy validity	Every strategy is competitive, cooperative or balanced.

Across the 690 official runs, task-state, reputation-bound and strategy-validity reporters pass in all records. Maximum absolute energy and reward residuals are 2.96×10^{-12} and 2.73×10^{-12} , respectively. These residuals are consistent with floating-point arithmetic rather than accounting failure. Raw exports are retained as immutable evidence; the per-step noncanonical BS01 export is excluded from the official 690-row analysis and retained only for traceability.

3.1 Internal validity controls

- Paired seeds expose each condition in an experiment to comparable stochastic initialization and agent-order effects.
- Stopping conditions are identical within each experiment: no active task remains, the mission time limit is reached, or all robots are depleted.

- BehaviorSpace view and per-step metric output are disabled in official experiments to avoid unnecessary execution variation.
- Raw files, clean experiment files, the harmonized master dataset, analysis notebooks and checksums form an explicit evidence chain.
- The same baseline condition is reproduced across multiple experiments, providing a cross-file consistency check.

4. Experimental Design

The official programme contains seven BehaviorSpace experiments and 690 run-level observations. Each condition uses seeds 1 through 30. Seed acts as a matched block, allowing comparisons to subtract much of the initialization noise and focus on the manipulated mechanism. Cross-experiment comparisons are descriptive because experiments were not jointly randomized as one factorial design.

Table 7. BehaviorSpace experimental programme.

ID	Experiment	Conditions	Runs	Primary analytical question
BS01	Baseline replication	1	30	Is decentralized coordination operationally viable under the validated baseline?
BS02	Reward regime comparison	4	120	Do reward rules alter completion, inequality or auction friction?
BS03	Communication degradation	2 x 2	120	Is communication reach or temporary reliability the stronger constraint?
BS04	Capability heterogeneity	3	90	Does capability diversity have an intermediate optimum?
BS05	Cooperative task demand	3	90	Does coalition demand create a nonlinear capacity threshold?
BS06	Reputation x adaptation	2 x 2	120	Do reliability signals and learning improve mission performance jointly?
BS07	Mission incentive strength	4	120	Do larger mission bonuses create performance gains proportional to cost?
Total			690	

4.1 Validated baseline

The validated baseline uses 15 robots, 30 tasks, capability heterogeneity 0.50, energy heterogeneity 0.30, mission time limit 500, movement cost rate 0.10, communication radius 8, communication quality threshold 0.25, communication failure rate 0.05, cooperative-task proportion 0.40, maximum coalition size 3, equal-share rewards, reputation enabled, adaptive strategies enabled and mixed initial strategies. The complete parameter list is provided in Appendix A.

4.2 Hypothesis register

The seven hypotheses consolidate the mechanism questions represented by BS01-BS07. The paired designs and experiment contrasts were documented in the project materials. The final seven-hypothesis register, the H1 90% benchmark and the integrated interpretation were formalized during post-run synthesis. The analyses are therefore structured and transparent but should not be represented as fully preregistered confirmatory tests.

5. Data and Statistical Methodology

5.1 Canonical dataset

The canonical master dataset contains 690 records and 49 harmonized variables. It contains no exact duplicate rows and no duplicate Experiment-Condition-Seed keys. Seven clean experiment files support within-experiment analysis, while the master file supports consistent names and integrated descriptive comparisons. Raw exports remain unmodified.

5.2 Paired-seed design

For a given experiment, each seed represents the same stochastic block across conditions. Paired or repeated-measures analyses preserve this structure. Independent-sample tests would discard the block design and inflate unexplained variance. For BS03 and BS06, 2 x 2 main effects and interactions are computed within seed before testing contrasts against zero.

Table 8. Statistical procedures and reporting rules.

Purpose	Procedure	Reported evidence
Baseline benchmark	One-sample Wilcoxon signed-rank; bootstrap mean interval	W statistic, one-sided p-value, rank-biserial effect and 95% bootstrap interval.
Three- or four-condition comparison	Friedman repeated-measures test	Chi-square, degrees of freedom, p-value and Kendall's W.
Planned paired comparison	Wilcoxon signed-rank	Mean and median paired difference, rank-biserial effect, raw and Holm-adjusted p-values.
2 x 2 mechanisms	Within-seed main-effect and interaction contrasts	Paired contrast magnitude, rank-biserial effect and Holm-adjusted p-value.
Condition uncertainty	Percentile bootstrap of the mean	95% interval shown in figures and condition tables.
Decision rule	Alpha 0.05 with family-level Holm correction	Primary-outcome decisions separated from secondary process evidence.

5.3 Interpretation discipline

Statistical non-significance is not treated as evidence of equivalence. Process effects are not automatically interpreted as mission effects. P-values are reported together with paired effect magnitudes, condition means and mechanism-specific diagnostics. Cross-experiment rankings use percentage-point differences as descriptive evidence and do not imply a single common randomized treatment scale.

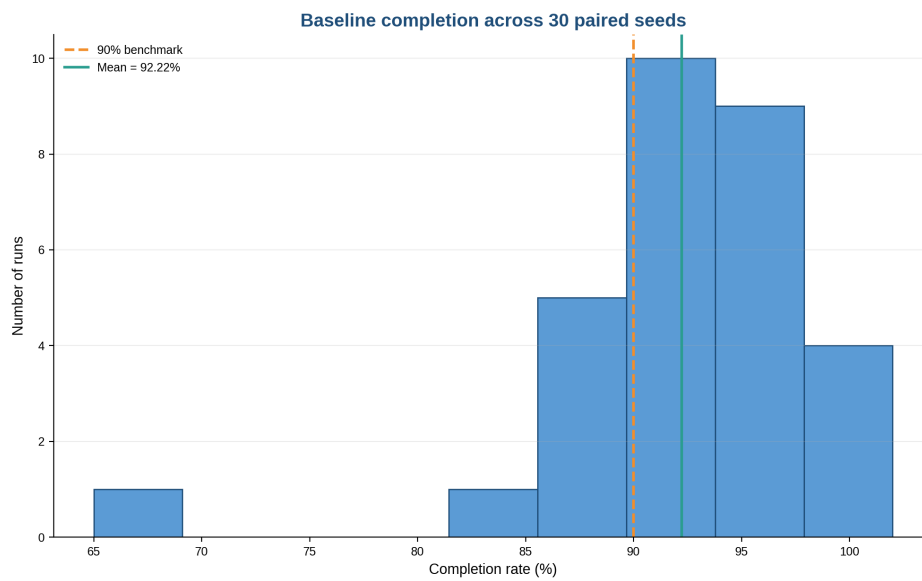
6. Hypothesis-Based Results

6.1 H1 - Baseline mission viability

Hypothesis
Under the validated baseline, decentralized coordination will achieve task completion above the 90% operational benchmark while recorded invariants remain valid.

Table 9. Baseline performance summary.

Metric	Mean	SD	Median	Interpretation
Completion rate	92.22%	6.80 pp	93.33%	23 of 30 runs reached at least 90%; four reached 100%.
Information coverage	95.16%	3.34 pp	96.11%	Most task identifiers entered the distributed knowledge network.
Coalition success	78.24%	12.17 pp	78.17%	Conditional success once a coalition was formed.
Coalition formation efficiency	4.79%	-	-	Feasible coalitions were rare relative to formation attempts.
Mission efficiency	1.482	0.541	1.538	Comparative task-value return on energy and coordination activity.
Reward Gini	0.296	0.071	0.289	Moderate reward concentration under equal-share.

**Figure 3. Baseline completion distribution across 30 paired seeds. The dashed line marks the 90% benchmark.**

Mean completion is 27.67 of 30 tasks, or 92.22%. A one-sided Wilcoxon signed-rank test of run-level completion against 90% gives $W = 231$, $p = 0.0098$ and rank-biserial = 0.540. This supports a typical-run advantage over the benchmark. The bootstrap 95% interval for the arithmetic mean is [89.67%, 94.44%], which crosses 90%. These statistics answer different questions: Wilcoxon evaluates the signed distribution of run-level deviations, whereas the bootstrap interval quantifies uncertainty around the mean. H1 is supported with caution rather than as an unqualified mean-above-90% claim.

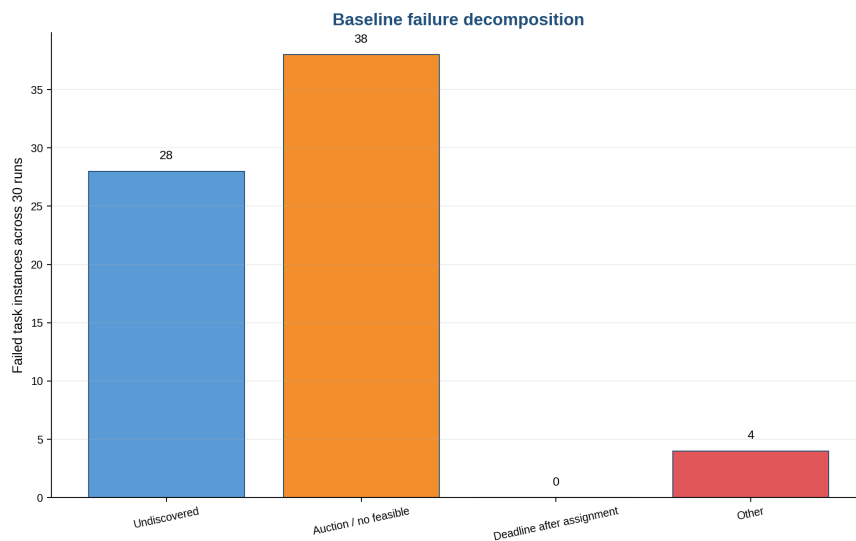


Figure 4. Baseline failure decomposition across 30 runs. Most failures occurred before successful assignment or execution.

Across the 30 runs, 38 of 70 failures (54.29%) are auction/no-feasible outcomes, 28 (40.00%) are undiscovered, none are deadline-after-assignment failures and four (5.71%) are other. The baseline shortfall therefore arises primarily before execution. Improving motion or task service alone would not address most failures.

Decision

H1 supported with caution. The baseline is operationally strong and internally consistent, but stochastic low-completion runs remain and the mean confidence interval overlaps the benchmark.

6.2 H2 - Reward-regime trade-off

Hypothesis

Leader-priority reward allocation will increase inequality and coordination friction relative to equal-share without improving task completion.

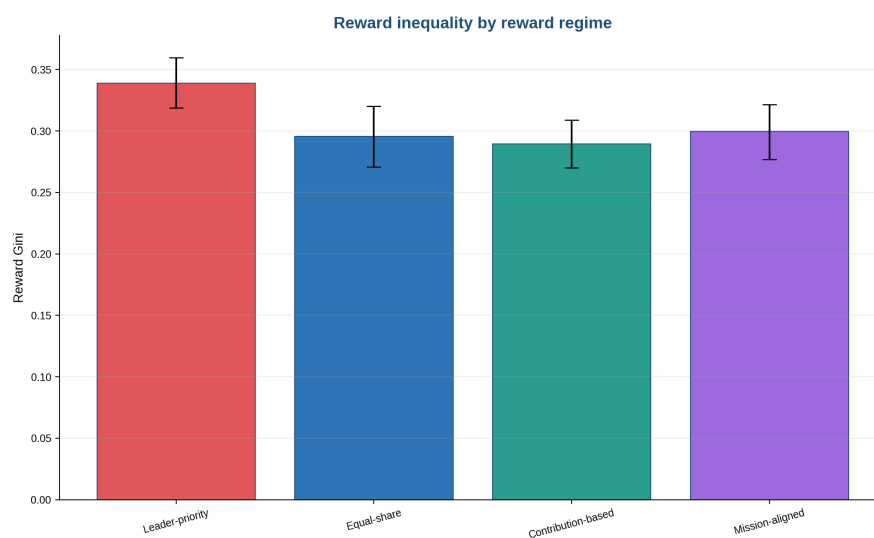


Figure 5. Reward inequality by reward regime. Error bars show bootstrap 95% intervals for condition means.

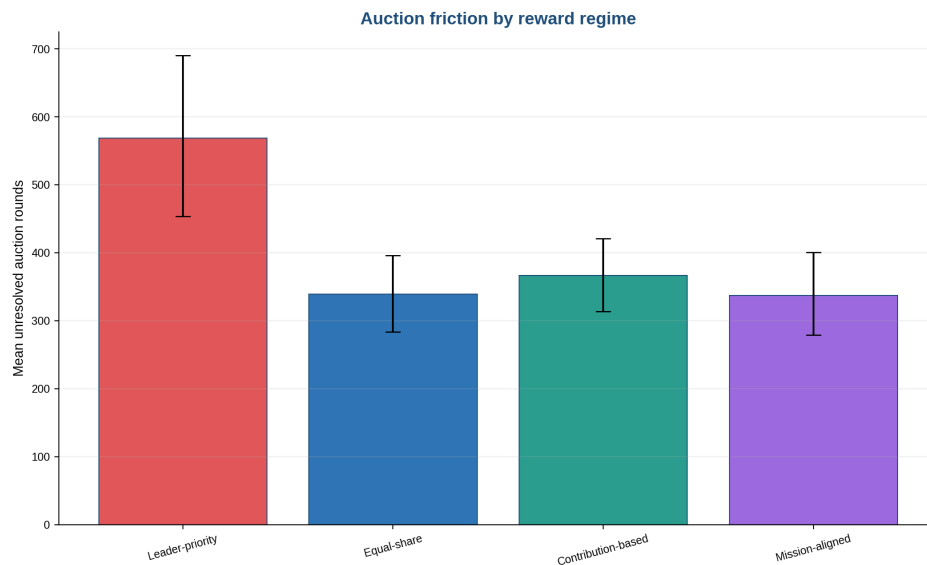


Figure 6. Auction friction by reward regime. Leader-priority creates substantially more unresolved auction rounds.

Table 10. Reward-regime inferential results.

Outcome	Global result	Planned leader-priority vs equal-share	Interpretation
Completion rate	Friedman chi-square(3) = 0.996; p = 0.802; W = 0.011	Mean difference -0.11 pp; Holm p = 1.000	No completion advantage from reward regime.
Reward Gini	Chi-square(3) = 8.939; p = 0.030; W = 0.099	+0.0432; Holm p = 0.0226; rrb = 0.523	Leader-priority increases inequality by about 14.6% relative to equal-share.
Unresolved auctions	Chi-square(3) = 27.763; p = 4.07e-06; W = 0.308	+229.63 rounds; Holm p = 4.2e-05; rrb = 0.828	Large increase in coordination congestion.
Coordination overhead	Global p = 0.051	+20.10 units; Holm p = 0.0199; rrb = 0.557	Planned comparison identifies additional process cost.

Leader-priority produces the highest mean Reward Gini (0.339), compared with 0.296 for equal-share, 0.289 for contribution-based and 0.300 for mission-aligned. Mean unresolved auction rounds rise from 338.80 under equal-share to 568.43 under leader-priority, a 67.8% increase. Completion remains statistically indistinguishable. The mechanism changes who is rewarded and how congested allocation becomes, not how many tasks are ultimately completed.

Decision

H2 supported. Equal-share is the defensible reference regime because it preserves completion with lower inequality and friction. Contribution-based rewards are a viable fairness alternative when contribution can be measured credibly.

6.3 H3 - Communication structure

Hypothesis

Structural communication reach will have a larger beneficial effect on mission outcomes than a moderate increase in temporary link failure.

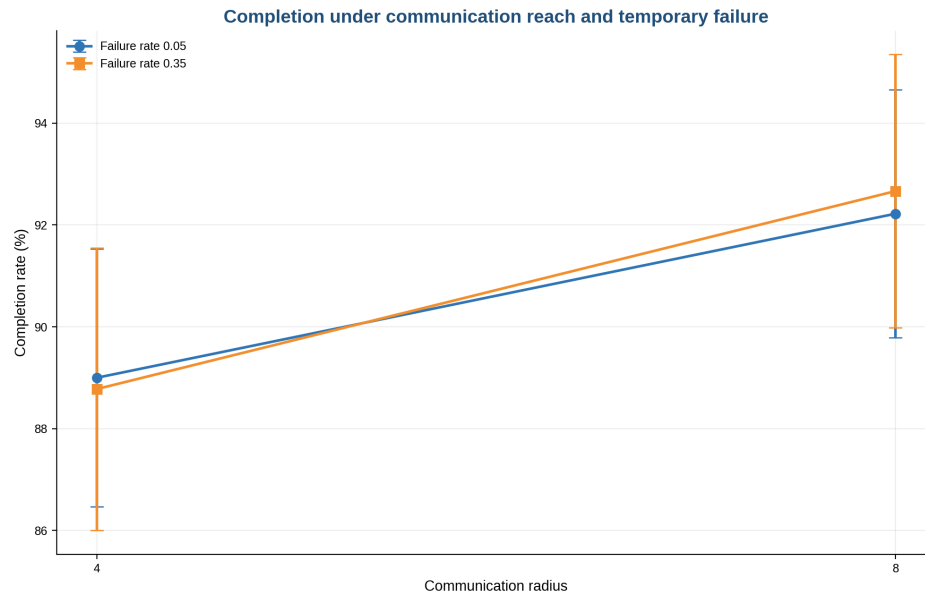


Figure 7. Completion under communication reach and temporary link failure. Error bars show approximate 95% intervals.

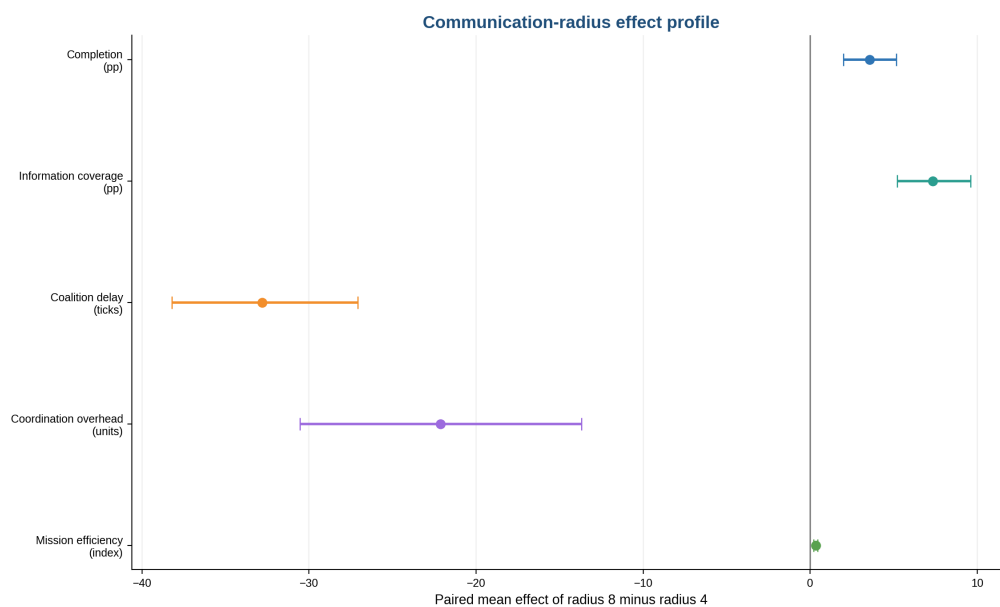


Figure 8. Paired effect profile of increasing communication radius from 4 to 8. Negative delay and overhead effects are beneficial.

Table 11. Communication-radius paired effects.

Outcome	Mean radius effect	Holm p	Rank-biserial	Interpretation
Completion rate	+3.56 pp	0.000663	0.826	Broader reach increases completed tasks.
Information coverage	+7.36 pp	3.07e-07	0.953	Task identifiers reach a larger share of robots.
Mean coalition delay	-32.80 ticks	2.79e-08	-0.987	Feasible partners are assembled substantially faster.
Coordination overhead	-22.12 units	0.000189	-0.781	Fewer repeated coordination actions per completed task.
Mission efficiency	+0.332	2.77e-05	0.845	Higher task-value return on energy and coordination activity.

Raising link failure from 0.05 to 0.35 does not produce a reliable completion effect: mean contrast +0.11 points, Holm $p = 1.000$. Radius-by-failure interactions are not significant. Within the tested range, the system tolerates moderate temporary loss when the structural network remains sufficiently connected. Radius affects both knowledge diffusion and the reachable partner set; this explains the coherent benefits across coverage, formation delay, overhead, completion and efficiency.

Decision

H3 supported. Communication reach is a mission resource, not merely a messaging parameter. Structural connectivity dominates moderate temporary failure in the tested design.

6.4 H4 - Capability heterogeneity optimum

Hypothesis

Moderate capability heterogeneity (0.5) will outperform both homogeneous (0.0) and highly heterogeneous (1.0) teams.

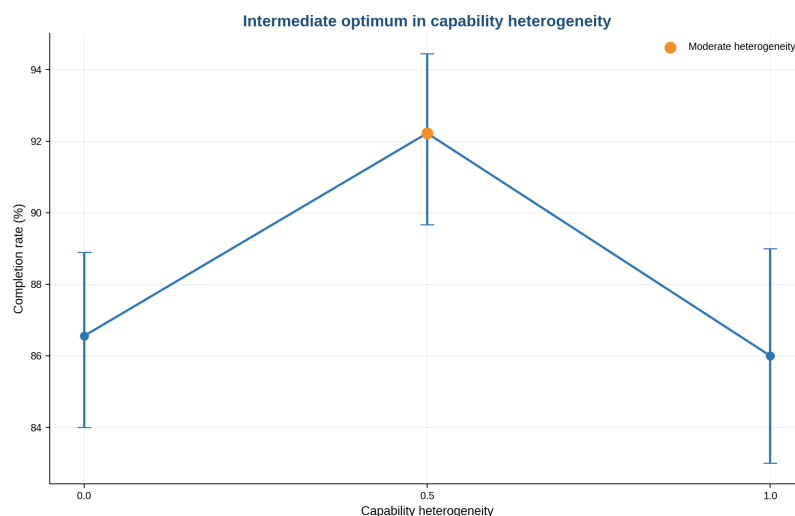


Figure 9. Intermediate optimum in capability heterogeneity. Error bars show bootstrap 95% intervals.

Table 12. Capability-heterogeneity results.

Outcome	0.0	0.5	1.0	Planned moderate-vs-extremes result
Completion rate	86.56%	92.22%	86.00%	+5.94 pp; $W = 375$; $p = 4.46e-05$
Mission efficiency	Lower	Highest	Lower	+0.540; $p = 1.93e-07$
Global completion test				Friedman chi-square(2) = 12.355; $p = 0.0021$; Kendall $W = 0.206$

The moderate condition exceeds both extremes after Holm correction. Homogeneous teams have high information coverage but weak complementarity and intense repeated bidding. Highly heterogeneous teams introduce specialization and matching fragmentation, lowering coverage, increasing coalition delay, releases and reward inequality. Moderate heterogeneity provides enough complementarity to expand feasible combinations without making suitable partners too rare.

Decision

H4 supported. More diversity is not monotonically better; team composition should optimize mission-relevant complementarity and redundancy rather than maximize dispersion.

6.5 H5 - Cooperative-demand threshold

Hypothesis

Cooperative-task demand will create a nonlinear coordination-capacity threshold, with the 0.8 condition causing a disproportionately larger completion decline than the 0.4 condition.

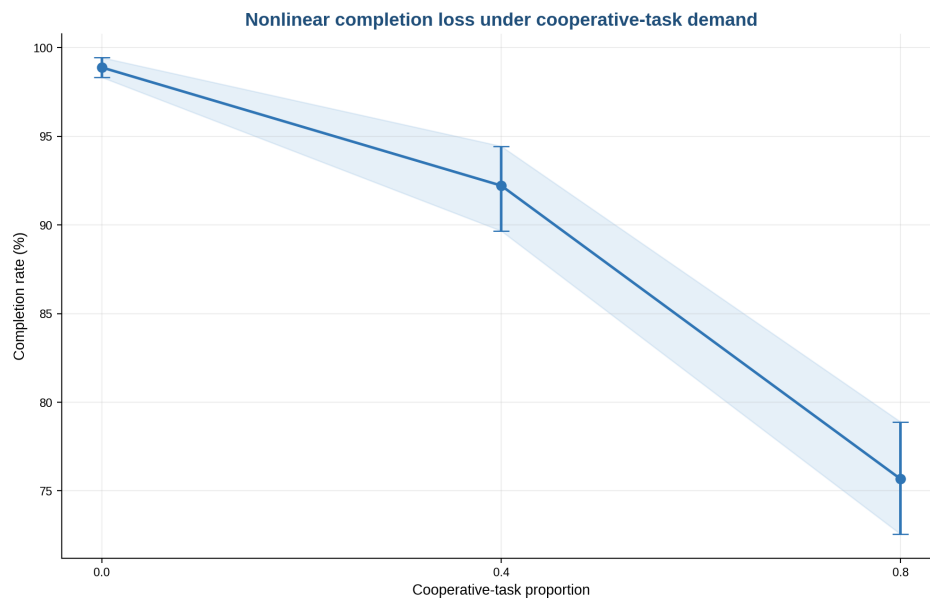


Figure 10. Nonlinear completion loss under cooperative-task demand. Error bars show bootstrap 95% intervals.

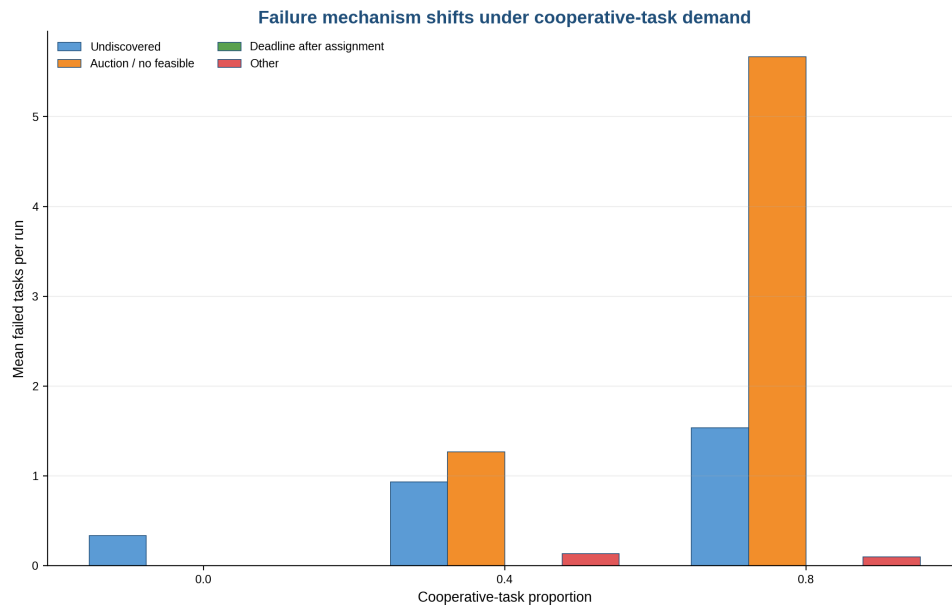


Figure 11. Failure mechanism shifts under cooperative-task demand. Auction/no-feasible failures dominate at high demand.

Table 13. Cooperative-demand threshold results.

Outcome	0.0	0.4	0.8	Planned nonlinear evidence
Completion rate	98.89%	92.22%	75.67%	Second-difference +9.89 pp; $p = 0.00142$; $rrb = 0.634$
Auction/no-feasible failures	0.00	1.27	5.67	Nonlinear contrast +3.13; $p = 0.000296$; $rrb = 0.716$
Formation efficiency	Higher	Baseline	Lower	$0.8 - 0.4 = -1.28$ points; $p = 0.000313$
Mean coalition delay	Lower	Baseline	Higher	$0.8 - 0.4 = +11.51$ ticks; $p = 0.00578$

Completion falls by 6.67 points from 0.0 to 0.4 and by 16.56 points from 0.4 to 0.8; the second decline is 2.48 times larger. At the 0.8 condition, 170 of 219 failed tasks (77.6%) are auction/no-feasible failures. Coalition success remains near 79%, showing that formed coalitions generally execute successfully. The system fails because it cannot form enough feasible coalitions before deadlines.

Decision

H5 supported. The dominant threshold is coalition-formation capacity, not the reliability of coalitions after they have formed.

6.6 H6 - Reputation and adaptation synergy

Hypothesis

Reputation and adaptive strategies will improve mission outcomes most strongly when jointly enabled.

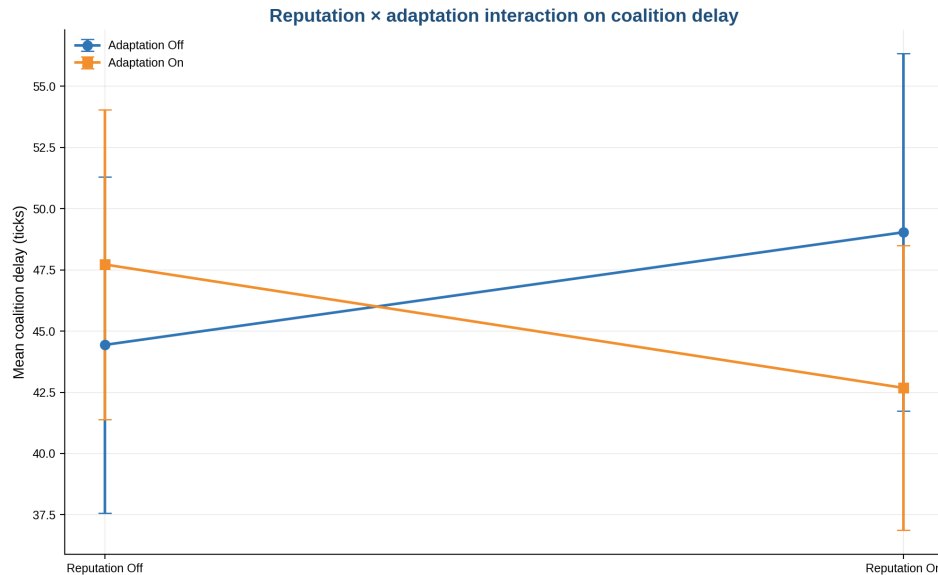


Figure 12. Reputation-by-adaptation interaction on coalition delay. Joint activation produces a process-level reduction.

Table 14. Reputation and adaptation factorial results.

Outcome	Contrast	Mean effect	Holm p	Interpretation
Completion	Reputation main effect	+0.28 pp	1.000	No reliable mission-level benefit.
Completion	Adaptation main effect	+0.94 pp	0.998	No reliable mission-level benefit.
Completion	Interaction	+0.78 pp	1.000	Joint activation does not improve completion reliably.
Coalition delay	Reputation main effect	-0.22 ticks	1.000	No main effect.
Coalition delay	Adaptation main effect	-1.54 ticks	1.000	No main effect.
Coalition delay	Interaction	-9.65 ticks	0.0228	Joint activation reduces delay beyond additive expectation.

The mechanisms are active: adaptive conditions generate about 52 strategy switches per run and reputation-enabled conditions move mean reputation away from its 0.50 initialization. Activity is not equivalent to mission improvement. The significant interaction on delay suggests that learned strategies can exploit reliability information to form teams faster, but the process gain is insufficient or occurs too late to change final completion under the current baseline.

Decision

H6 is not supported for the primary outcome. The delay interaction is credible secondary process evidence and should be reported without upgrading it to mission-level validation.

6.7 H7 - Mission-incentive saturation

Hypothesis

Increasing mission-incentive strength will exhibit diminishing returns: payments will rise monotonically while task completion will not reliably improve.

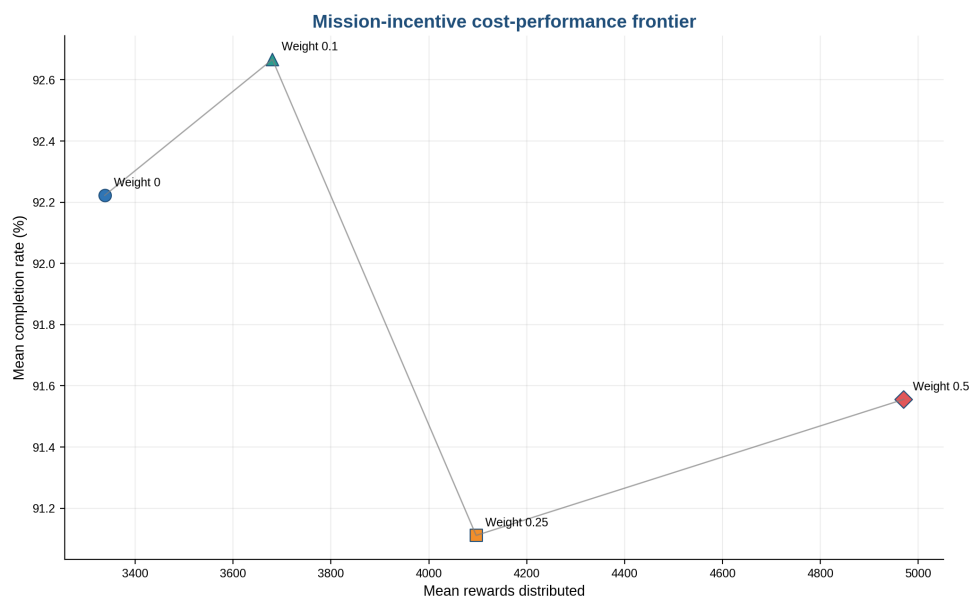


Figure 13. Mission-incentive cost-performance frontier. Weights 0.25 and 0.50 cost more and have lower mean completion than zero incentive.

Table 15. Mission-incentive cost-performance results.

Weight	Completion	Rewards distributed	Incremental reward	Coalition success	Cost-performance interpretation
0.00	92.22%	3,337.58	0.00	78.24%	Reference condition.
0.10	92.67%	3,680.33	+342.76	81.94%	Small descriptive completion gain; not statistically reliable.
0.25	91.11%	4,096.76	+759.18	82.00%	Dominated by zero on cost and mean completion.
0.50	91.56%	4,970.47	+1,632.90	82.68%	Dominated by zero on cost and mean completion.

Completion does not differ by incentive weight: Friedman chi-square(3) = 4.515, $p = 0.211$, Kendall W = 0.050. Rewards respond almost perfectly: chi-square(3) = 86.52, $p = 1.22\text{e-}18$, Kendall W = 0.961. Planned comparisons against zero show significant payout increases with rank-biserial effects from 0.996 to 1.000. Coalition success increases by approximately 3.70 to 4.44 points, each Holm $p = 0.0455$, but this conditional process gain does not increase total completion.

Decision

H7 supported. Incentive intensity is not the binding mechanism. A low 0.10 weight may remain an exploratory condition, but higher weights lack a cost-effectiveness justification in the tested architecture.

7. Integrated Analysis

7.1 Structural coordination capacity dominates

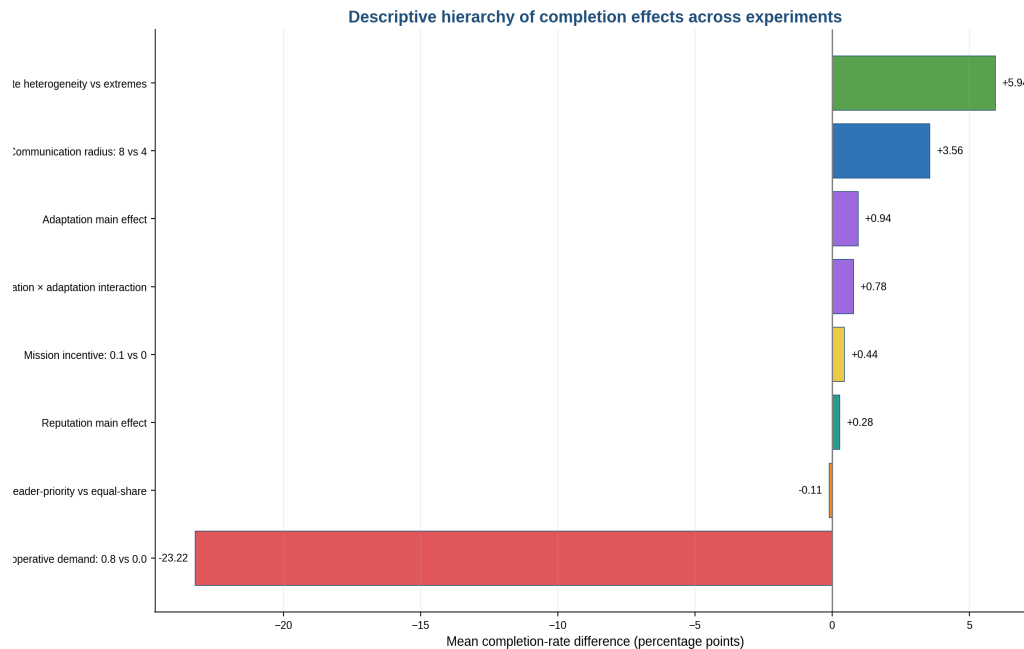


Figure 14. Descriptive hierarchy of completion effects across experiments. Values are mean percentage-point differences and are not a joint factorial estimate.

The effect hierarchy places cooperative demand, capability heterogeneity and communication reach above reward, reputation and adaptation mechanisms. This ordering is visible in absolute completion differences, paired effect sizes and coherent movement across process metrics. Cooperative-demand overload changes completion by -23.22 points. Moderate heterogeneity provides +5.94 points relative to the average of both extremes. Communication radius adds +3.56 points. By comparison, adaptation, reputation, their interaction and a low mission bonus each shift mean completion by less than one point.

Table 16. Integrated evidence hierarchy and design interpretation.

Evidence class	Strongest observation	Mechanism diagnosis	Design implication
Cooperative load	0.8 demand reduces completion by 23.22 pp	Formation capacity is exceeded; no-feasible allocation dominates.	Increase coalition-search capacity or limit simultaneous cooperative demand.
Team composition	0.5 heterogeneity outperforms both extremes by 5.94 pp	Complementarity has an optimum; extreme specialization fragments matching.	Engineer role-relevant diversity with redundancy.
Communication structure	Radius 8 improves all tested mission/process metrics	Reach expands both information diffusion and feasible partner access.	Use relays, repositioning or network-aware roles.
Reward policy	Leader-priority increases inequality and congestion	Distributional incentives can worsen allocation without affecting completion.	Use equal-share as the reference regime.

Evidence class	Strongest observation	Mechanism diagnosis	Design implication
Reputation and learning	Joint activation reduces delay, not completion	Process improvement remains below the mission bottleneck.	Add persistent reliability differences and more informative experience.
Mission bonuses	Payouts rise monotonically; completion does not	Incentives cannot purchase missing structural feasibility.	Evaluate bonuses through explicit cost-effectiveness criteria.

7.2 Information coverage is necessary but not sufficient

High coverage does not guarantee high completion. Homogeneous capability teams reach 97.39% information coverage but complete only 86.56% of tasks. Awareness without compatible capability combinations produces congestion rather than execution. Moderate heterogeneity can have lower coverage and higher completion because the reachable agent set contains more useful complementary capability profiles.

7.3 Conditional success can conceal formation failure

Coalition success is conditioned on a coalition having formed. At high cooperative demand, success remains near 79%, yet overall completion collapses because formation efficiency falls and auction/no-feasible failures increase. A system can therefore report healthy execution among selected teams while failing at mission scale. Formation attempts, formation efficiency, no-feasible failures and delay are necessary diagnostic metrics.

7.4 Fairness and performance are separable

Equal-share and contribution-based rewards provide similar completion but different inequality. This allows reward policy to be selected based on fairness, interpretability and measurement credibility rather than unsupported performance claims. Leader-priority is unattractive on both distribution and congestion grounds. Mission-aligned rewards require an explicit budget rationale because additional payment does not produce a reliable mission benefit.

7.5 Process improvement is not automatically mission improvement

H6 and H7 improve process outcomes without moving completion. Joint reputation and adaptation reduce coalition delay; bonuses modestly improve coalition success. Another stage remains binding. This stage-coupling interpretation is more informative than labeling the mechanisms ineffective. Future experiments should vary deadline pressure, task load, repeated-mission reliability and learning duration to determine when process gains become mission gains.

8. Engineering and Research Implications

8.1 Redesign coalition formation

The highest-priority modification is targeted partner discovery and feasible coalition search. Candidate approaches include capability indexing, distributed constraint satisfaction, bounded combinatorial search, role-based partner requests and memory of previously unsuccessful combinations. A replacement should be evaluated on formation efficiency, delay, unresolved rounds, computational overhead and robustness under high cooperative demand.

8.2 Treat communication reach as a controllable resource

The broad radius effect implies that network structure should become part of mission behavior. Relay roles, temporary communication positioning, task-priority routing and connection-aware movement may create reach without assuming uniformly larger physical range. Future models should separate range, routing, bandwidth, latency, loss and message priority.

8.3 Engineer mission-relevant heterogeneity

Team composition should maximize coverage of task-relevant capability combinations while preserving redundancy and partner availability. A scalar heterogeneity parameter is only a first approximation. Future designs should vary capability correlations, rarity, redundancy and task-requirement distributions to identify resilient compositions.

8.4 Maintain equal-share as the reference reward regime

Equal-share is transparent, reproducible and less congested than leader-priority. Contribution-based rewards are defensible when contribution can be measured without distortion. Mission bonuses should begin at zero or a low exploratory value and should be judged against completion, coalition formation and explicit reward budget, not coalition success alone.

8.5 Make reputation informative

Reputation becomes useful when robots have persistent differences in reliability, sensing quality, energy efficiency or failure propensity across repeated missions. The current model has limited stable signal. Future experiments should introduce controlled robot-specific reliability and evaluate whether reputation becomes calibrated, discriminative and operationally useful.

9. Limitations and Threats to Validity

9.1 Construct validity

MOSAIC represents high-level coordination, not physical robotics. Energy, reward, capability, reputation, overhead and efficiency are dimensionless. Task knowledge is an identifier rather than a semantic representation. Coalition contribution is a simplified average, and execution progress is equal per member once the team is feasible. These abstractions are appropriate for mechanism exploration but restrict direct interpretation.

9.2 Internal validity

The observer schedules all processes and NetLogo randomizes agent order inside ask blocks. The toroidal world removes boundaries and can change encounter structure. Communication knowledge never decays. The greedy coalition heuristic may miss feasible combinations. Expected leader reward assumes the discoverer receives the leader share, while actual leadership may be assigned differently. These limitations can generate mechanism-specific biases that should be tested in future versions.

9.3 Statistical conclusion validity

Thirty paired seeds provide efficient comparisons but do not exhaust stochastic uncertainty. Holm correction is applied within planned families, not across every test in the report. Some variables are bounded, tied or zero-inflated. H1 uses a post-run operational benchmark. Cross-experiment effect ordering is descriptive and should not be treated as a common causal scale from a single experiment.

9.4 External validity

There is no empirical calibration, GIS input, hardware-in-the-loop execution, collision avoidance, detailed radio model, dynamic task arrival, adversarial behavior, strategic defection or human intervention. Results are not safety certification, economic valuation or field forecasts. External validity requires staged calibration, broader sensitivity analysis, hardware-aware simulation and physical trials.

10. Reproducibility and Artifact Availability

Reproduction requires NetLogo 7.0.4, the MOSAIC v0.4.0 model, the documented baseline or experiment parameters, seeds 1 through 30, the official stopping condition and no manual state intervention. The prepared pre-publication artifact set includes the executable model, standalone ODD Protocol, User Manual, experimental design, raw BehaviorSpace exports, seven clean experiment files, the 690-row master dataset, analysis notebooks, statistical tables, figures, manifest and checksums.

The canonical analysis file is processed/MOSAIC_All_Experiments_Master_Runs.csv. Analyses must preserve the original master dataset and write derived outputs separately. The Experiment-Condition-Seed key and paired seed variable must remain intact. The excluded noncanonical per-step BS01 export is retained only for traceability and must not be combined with the official 690 run-level records.

11. Conclusion

MOSAIC v0.4.0 provides a reproducible computational platform for studying decentralized mission coordination among heterogeneous robots. The ODD-structured specification makes the implemented assumptions, entities, schedule, design concepts and submodels explicit. The 690-run evaluation provides a coherent mechanism diagnosis rather than a collection of isolated outcomes.

Communication reach, capability complementarity and coalition-formation capacity determine whether local competition becomes effective mission coordination. Reward rules affect fairness and congestion, but stronger payments do not compensate for missing information or infeasible partner sets. Reputation and adaptation show a limited process-level benefit without a robust mission-level gain. The central development implication is precise: improve decentralized partner discovery and coalition construction before increasing incentive intensity.

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Appendix A - Validated Baseline Parameters

Table 17. Validated baseline parameter values.

Parameter	Value	Parameter	Value
number-of-robots	15	number-of-tasks	30
robot-initial-energy	150	mission-time-limit	500
energy-heterogeneity	0.30	capability-heterogeneity	0.50
movement-cost-rate	0.10	detection-radius	2.50
communication-radius	8	communication-quality-threshold	0.25
communication-failure-rate	0.05	information-sharing-probability	0.90
coalition-task-proportion	0.40	maximum-coalition-size	3
reward-regime	equal-share	reputation-enabled?	true
adaptive-strategies?	true	initial-strategy	mixed
random-seed-value	42 for manual baseline	BehaviorSpace seeds	1-30

Generation parameters must be set before SETUP. Changing them after initialization does not regenerate the existing environment, robots or tasks. Mid-run parameter changes should be documented as interventions rather than baseline replications.

Appendix B - Exact Hypothesis Statistics

Exact tests, effect sizes, adjusted p-values, condition summaries and figure data are archived in the accompanying analysis tables and in MOSAIC_v0.4.0_Hypothesis_Analyses_Executed.ipynb. The executed notebook is the computational record for the inferential claims in Section 6.

- H1: $W = 231$; $p = 0.009813$; rank-biserial = 0.540; bootstrap mean interval [89.67%, 94.44%].
- H2 unresolved auctions: Friedman chi-square = 27.763; $p = 4.07e-06$; leader-priority vs equal-share Holm $p = 4.2e-05$.
- H3 communication radius completion effect: +3.56 pp; Holm $p = 0.000663$; rank-biserial = 0.826.
- H4 moderate-vs-extremes completion contrast: +5.94 pp; $p = 4.46e-05$.
- H5 nonlinear completion contrast: +9.89 pp; $p = 0.001419$; rank-biserial = 0.634.
- H6 coalition-delay interaction: -9.65 ticks; Holm $p = 0.022836$; rank-biserial = -0.548.
- H7 rewards: Friedman chi-square = 86.52; $p = 1.22e-18$; Kendall $W = 0.961$.

Appendix C - Authorship and Institutional Context

MOSAIC was initially developed by Katherin Molina as a final project for the Introduction to Agent-Based Modeling course offered through Complexity Explorer of Santa Fe Institute and was subsequently extended in collaboration with Lorena Holguin, Software Engineer at Kahlo Group. Complexity Explorer and the Santa Fe Institute provided academic context only and do not own, sponsor, validate or endorse the model. Kahlo Group is listed only as Lorena Holguin's professional affiliation and did not fund, sponsor, supervise, validate or endorse the project.

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